Design of a Specialized Tokenizer for Mathematical Text

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October 10, 2026

Can Mathematical Tokenization Improve Model Capability?

Research question

Can we train tokenizers specifically for mathematical text that improve both **intrinsic tokenization** and **downstream mathematical capability**?

Question	What would count as evidence?
RQ1: tokenizer quality	Lossless compression, robustness, speed, etc.
RQ2: model capability	Mathematical tasks such as PDE/ODE solving.
Current status	Tokenizer experiments completed; downstream evaluation to follow.

Study Overview

Component	Summary
Dataset	We constructed and classified a dataset of mathematical formulas from multiple mathematical fields.
Models	We proposed two specialized mathematical tokenizers.
Baselines	We selected five tokenizers: classical BPE and Unigram methods, and tokenizers used by mature large language models.
Results	Our two tokenizers achieve leading performance on core intrinsic metrics and in the overall comparison.

Baseline Tokenizers

Category	Tokenizer	Actual ID space	Code / tokenizer source
Classical tokenizers	Byte BPE	1,024 / 2,048	huggingface/tokenizers
Classical tokenizers	SP Unigram	1,024 / 2,048	google/sentencepiece
General-Purpose Tokenizers	Qwen2.5	151,665	Qwen2.5-0.5B/tokenizer.json
General-Purpose Tokenizers	Llama 3	128,256	meta-llama/llama3/tokenizer.py
General-Purpose Tokenizers	o200k	200,019	openai/tiktoken

- Qwen2.5: used by the Qwen2.5 model family.
- Llama 3: used by the Meta Llama 3 model family.
- o200k_base: used by GPT-4o and GPT-4.1.

SP Unigram: Training and Inference

Training step	Procedure
Preparation	Use raw strings without language-specific word segmentation; represent spaces as a symbol, usually _.
1. Initialize	Create a large seed vocabulary of characters, frequent substrings and symbols; assign initial piece probabilities.
2. EM iterations	E: sum over possible segmentations to obtain expected piece counts. M: re-estimate probabilities from those counts.
3. Prune	Remove pieces with low contribution to corpus likelihood; repeat EM and pruning to reach 1,024 or 2,048 IDs.
4. Final model	A piece vocabulary and probabilities; an optional byte fallback table.

Inference

Build a segmentation lattice; for abc: $a+b+c$, $ab+c$, $a+bc$, or abc.

$$P(s) = \prod_i p(s_i), \qquad s^* = \arg\max_{s \in \mathcal{S}(x)} P(s).$$

Viterbi dynamic programming returns the most probable token sequence.

SP Unigram: A Two-Round EM Example

Setup: one training string ab; pieces a, b, ab; initially $p(a) = p(b) = p(ab) = 1/3$.

E step: segmentation likelihoods and posteriors

Round	(a, b) likelihood	ab likelihood	$P(x)$	Posterior: (a, b) / ab
1	$1/3 \times 1/3 = 1/9$	$1/3 = 3/9$	$4/9$	$1/4 = 0.25$ / $3/4 = 0.75$
2	$0.2 \times 0.2 = 0.04$	0.6	0.64	0.0625 / 0.9375

M step: expected counts and normalized probabilities

Round	$\mathbb{E}[n_a], \mathbb{E}[n_b], \mathbb{E}[n_{ab}]$	Total	$p(a), p(b), p(ab)$
1	$0.25,\ 0.25,\ 0.75$	1.25	$0.2,\ 0.2,\ 0.6$
2	$0.0625,\ 0.0625,\ 0.9375$	1.0625	$1/17,\ 1/17,\ 15/17$ $\approx 0.0588,\ 0.0588,\ 0.8824$

$$P(x) = \sum_{s \in \mathcal{S}(x)} \prod_i p(s_i), \qquad p(v) = \frac{\mathbb{E}[n_v]}{\sum_u \mathbb{E}[n_u]}.$$

For this one-string example, $p(ab) \to 1$ and $p(a), p(b) \to 0$: EM favors the whole piece ab. Low-contribution pieces become pruning candidates; required characters remain protected.

Our Two Routes: Contribution and Prior Art

Dimension	Math-atom BPE	Boundary AST
Vocabulary	Fit-derived complete-command/Unicode atoms; original BPE.	Exactly the same byte vocabulary and ID budget.
Segmentation	Original learned BPE merge order.	Minimum additive token/tree/boundary cost on a legal source graph.
Mathematical prior	Lexical units observed in training.	Lexical units plus a restricted source parser.
Deployment	Native reversible codec + original Rust BPE.	Fused native trie traversal and dynamic programming.
Novelty boundary	A reproducible domain-specific training recipe.	A tested structural trade-off; shortest paths and AST ideas are not new.

Central comparison

Holding the dictionary fixed isolates the effect of the structural segmentation rule. Comparing against generic tokenizers instead measures the complete specialized system.

Bai et al. (2026) ; Schmidt et al. (2024) ; Scarlatos and Lan (2023)

Corpus Sources and Their Roles

Source	Accepted formulas	Provenance / role
arXiv, fixed pre-2020 versions	51,229	CC BY / CC0; 283 contributing source papers; historical human-origin inference.
OpenAI math repository	2,810	Apache-2.0; conservatively labeled possible_mixed.
FunGrim	1,889	MIT; formula library, origin unknown.
SymPy	72	BSD-3-Clause; formula fixtures, origin unknown.
EqGPT structures / AI templates	Supplement only	Not counted as complete research-level LaTeX formulas in the 56,000-string catalog.

Source-specific distinction

EqGPT reports 221 handbook PDE structures and an operator vocabulary. Its structure strings are useful context, but are not automatically valid complete LaTeX equations.

Record-level traceability

Each accepted string retains its source revision, license and location. Formula text is stored separately from this metadata; redistribution follows the upstream record license.

Xu et al. (2025) ; Gebru et al. (2021)

Construction: From Licensed Source to Formula Record

Step	Operation	Retained evidence
1. Acquire	Official arXiv URLs; cache first, one connection, at least 4 s between requests.	Version date, license URL, source archive hash.
2. Extract	Find inline/display/multirow math; bounded source-macro expansion.	Original LaTeX, member path and source span.
3. Filter	Reject prose, isolated labels, executable commands and damaged/trivial fragments.	Explicit rejection reason and quarantine record.
4. Validate	Independent KaTeX parse and source/expanded rendering fidelity.	Reference spans and rendered-equivalence hashes.
5. Partition	Exact/template/render-equivalent exclusion against earlier exposures.	Frozen formula IDs, source groups and split hashes.

No hidden mathematical rewriting

The encoder is evaluated on the accepted stored string. Formula normalization is a recorded dataset operation, not a model-specific test preprocessing step.

Gebru et al. (2021) ; Bai et al. (2026)

Taxonomy: Formula Type Is Not Paper Topic

Formula domain	Count	Formula domain	Count
analysis	20,233	geometry / topology	9,135
combinatorics	5,563	algebra	4,431
number / theory	4,273	mathematical / physics	3,850
optimization	3,211	probability	3,171
statistics	1,056	logic	662
pde	356	ode	46
linear / algebra	13		

Separate axes

Formula domain: strict PDE vs. other mathematics.
Paper domain: source subject, stored independently.
Types: integral, sum/product, matrix, differential relation and system.

PDE limitation

Only 356 strings meet the strict PDE rule. Other formulas from PDE papers retain their structural category; the catalog is not yet a large PDE-solver training set.

Provenance Labels and Two Complexity Classes

Origin label	Count	Interpretation
Historical human	51,229	Fixed pre-2020 manuscripts; inference from source history.
Possible mixed	2,810	Model-produced / edited manuscript sources; no per-formula attribution.
Unknown	1,961	Library provenance does not establish who typed each formula.
Fully AI generated	0 in main catalog	Synthetic templates remain separate supplements.

Fixed syntax-complexity score

$$s = .40\bar{L} + .20\bar{C} + .15\bar{D} + .15\bar{S} + .10\bar{O}, \quad \bar{z} = \min(z/z_0, 2).$$

L: lexeme length; C: command types; D: nesting; S: symbol types; O: operators. Normalizers: $(60, 6, 3, 18, 12)$. Complex iff $s \geq .65$.

Class balance

37,400 simple and 18,600 complex formulas. These are reproducible syntax strata, not theorem difficulty levels.

Frozen Partitions and Generalization Protocol

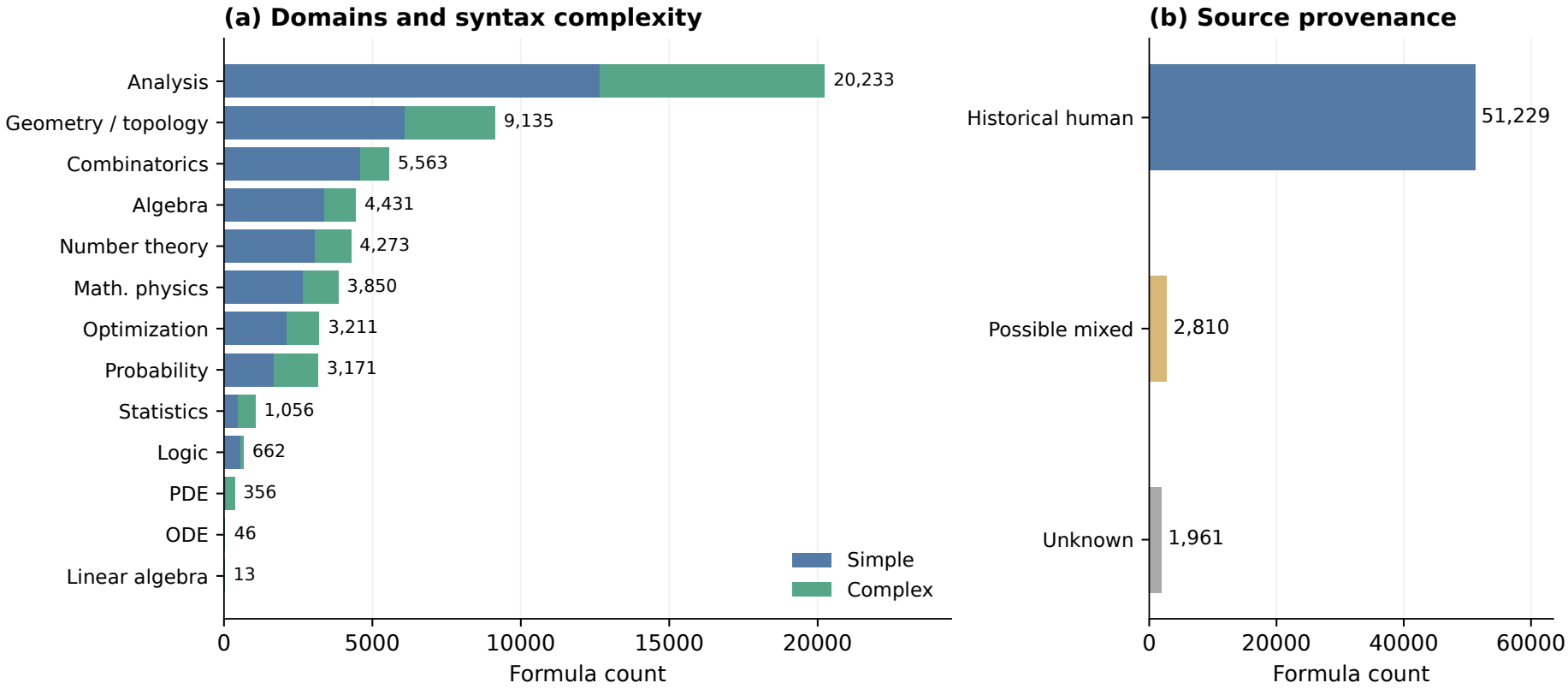
Partition / scenario	Formulas	Use
v6 train / dev	36,615 / 7,693	Source-separated pools; active scenario subsets below.
v6 new-paper test	3,465	Previously exposed; regression check only.
v7 new-paper / extreme	3,933 / 1,961	Previously exposed; regression and domain checks.
v8 new-paper / extreme	4,294 / 1,428	Final sealed source test, scored after model freeze.
Synthetic / long-input stress	521	Explicit stress diagnostics, outside the natural catalog.

- **Broad**: fit 30,100; auxiliary training/reward pool 512; dev 965.
- **Specialized**: fit 8,124; auxiliary pool 512; dev 249. Target and adjacent source categories excluded.
- Extreme subsets overlap their parent test. They are not additional independent samples.

Do not train on the whole catalog

56,000 is an organized library size. Its ID-only split manifests preserve the training/test boundary.

Corpus Composition and Coverage



What the composition implies

Analysis and geometry dominate. PDE, ODE and linear algebra are sparse. Report paper/domain/complexity strata and avoid interpreting an overall average as uniform mathematical coverage.

Route 1: Learning a Mathematical Lexical Alphabet

Training alphabet

$$A = \{\text{all 256 bytes}\} \cup \mathrm{Top}_{384}\{\text{observed complete commands / Unicode units}\}.$$

Atoms are derived from fit records only, ranked by frequency with deterministic tie rules. The actual primitive counts are 596 (broad) and 495 (specialized).

Stage	Implementation	Budget accounting
Reversible codec	Map observed lexical units to private-use symbols.	One primitive ID per atom; unknown strings retain byte coverage.
Vocabulary training	Original HF BPE on fit + the allowed 512-record auxiliary pool.	Total IDs capped at 1,024 or 2,048.
Inference	Native lexical codec, then original Rust BPE.	No AST path search and no policy network.

Why this route remains

It is the simple, fast mathematical reference against which extra structural machinery must justify itself. No manually enumerated command vocabulary is required.

Sennrich, Haddow and Birch (2016) ; Bai et al. (2026)

Route 1: Exact Source Fidelity and Its Limits

Raw-string contract

For every evaluated x, $\mathrm{decode}(\mathrm{encode}(x)) = x$. The ordered token byte payloads also concatenate to $\mathrm{UTF8}(x)$ exactly. No whitespace normalization is permitted.

Case	Behavior	What is not guaranteed
Seen command / Unicode atom	Codec retains the complete observed unit.	A larger token can still span an argument boundary.
Unseen command	Explicit byte coverage; no unknown-token deletion.	Every novel command need not be a single token.
Spacing / multiline source	Bytes are preserved exactly.	Layout variants need not receive identical token sequences.
Numbers / identifiers	BPE learns recurrent sequences.	Numeric magnitude is not encoded continuously.

Interpretation

High command integrity reflects observed alphabet coverage, not universal LaTeX grammar coverage or a learned theorem representation.

Golkar et al. (2023) ; Jung et al. (2024)

Route 2: Source Structure as a Segmentation Cost

Same vocabulary, different path

An edge (a, b) matches $x_{a:b} \in V$ at permitted lexical endpoints. The restricted parser supplies source spans and matched brace-content endpoints.

Additive objective

$$z^* = \arg \min_{z \in \mathcal{P}(x,V)} \left[|z| + \lambda \sum_{e \in z} c_T(e) + \mu \sum_{e \in z} c_B(e) \right].$$

c_T: token partially crosses a native source span.
c_B: number of brace-content endpoints strictly inside the token.
Development selection: $\lambda = .25$, $\mu = .30$.

A source-tree method

The parser is restricted and falls back to source syntax. It is not a full TeX interpreter or a semantic operator tree for arbitrary mathematics.

Schmidt et al. (2024) ; Scarlatos and Lan (2023)

Route 2: Exact Dynamic Programming on a Fixed Graph

Backward recurrence

$$D[n] = 0, \qquad D[a] = \min_{(a,b)\in E} \{1 + \lambda c_T(a,b) + \mu c_B(a,b) + D[b]\}.$$

A byte trie enumerates matching dictionary edges. A stored best endpoint reconstructs the token sequence. This is exact for the specified legal graph and additive objective.

Design	Mechanism	Reason
Lexical mask	Suppress cuts inside covered commands / codepoints.	Keep complete observed primitive units.
Tree cost	Endpoint sweeps detect partial span crossings.	Avoid testing every token against every span.
Boundary cost	Prefix sums count hidden brace-content endpoints.	Recover argument boundaries without hard blocking all compression.

Scope of optimality

The result is not globally optimal vocabulary learning, and a low source-tree cost does not imply low task loss. The shortest-path algorithm itself is established.

Schmidt et al. (2024) ; Uzan et al. (2024)

Route 2: Development Selection and Runtime Cost

Setting	Values / behavior
Prespecified grid	$\lambda \in \{0, .25, .75\}$; $\mu \in \{0, .05, .15, .30\}$: 12 candidates.
Selection criterion	Common old-dev utility plus .04 times missing parameter-boundary recall.
Feasibility guard	Compression regression at most 1%; boundary recall loss at most 1 percentage point vs. math-atom BPE.
Selected setting	All four budget/scenario conditions selected .25 / .30 before v8 scoring.
Public encoder	C frontend + trie + DP; all source parsing is included in throughput.

Engineering and trade-offs

Typical lattice work scales with input bytes, maximum piece length and source spans; the dense byte trie uses more memory than a compact sparse trie.
Compared with the older length-first path, the boundary term costs some speed but restores much of the parameter-boundary recall.

Deployment contract

A frozen dictionary and weights define deterministic IDs. Unknown lexical units use byte coverage; supported source parsing has an explicit one-million-byte input limit.

Exactly Two Routes and Five Baselines

System	Vocabulary / training	Role
Boundary AST (ours)	Common math-atom BPE vocabulary; fixed source-path objective.	Structure-aware route.
Math-atom BPE (ours)	Fit-derived alphabet; original BPE on allowed data.	Lexical domain route.
Byte BPE	Original HF byte BPE; same allowed training records.	Equal-capacity classical control.
SP Unigram	Original SentencePiece; identity normalization, byte fallback.	Equal-capacity probabilistic control.
Qwen2.5	Unmodified full official vocabulary.	General deployment reference.
Llama 3	Unmodified Meta reference tokenizer.	General deployment reference.
o200k	Original tiktoken encoding.	General deployment reference.

Current repository contract

Seven method names, 19 frozen configurations. Other models and experiments live only in the historical archive and cannot be selected by the active loader.

Common Inputs, Training Information and Timing

Axis	Controlled implementation
Vocabulary budgets	1,024 / 2,048 for trained methods; all primitive and reserved IDs count.
Allowed data	Identical scenario fit/auxiliary pools; generic trainers receive their union.
Test input	Original accepted raw strings; no per-model filtering, truncation or normalization.
Probability calibration	Same fit IDs for each scenario; mature models recalibrated for broad and specialized separately.
Timing	Seven shuffled/interleaved full public encode/decode runs; median throughput; caches cleared.
Statistical unit	Paper groups, not individual formulas copied across seeds.

Remaining asymmetry

Historical tuning counts and training wall time differ. Mature vocabulary budgets differ. These differences are disclosed rather than described as fully equal compute.

Efficiency, Fidelity and Information Measures

Metric	Definition / interpretation
Paper-macro tokens/byte ↓	$\frac{1}{P}\sum_p T_p/B_p$; prevents long papers from dominating the primary compression average.
Length distribution	Tokens/formula and 50th/95th/99th percentiles; inspect long-tail inflation.
Exact fidelity	Both decoded strings and concatenated raw token payloads equal the source.
Encode / decode speed	Raw UTF-8 MiB/s through the complete public interface.
Fit unigram BPB ↓	$-B^{-1}\sum_i \log_2[(c_i+\alpha)/(N+\alpha\lvert V\rvert)]$; fit-only count model, not LM perplexity.
Information / utilization	Renyi efficiency at $\alpha = 1, 2, 2.5, 3$; used IDs / actual vocabulary.

Sensitivity

Report smoothing at .01 / .1 / .5 / 1. Information efficiency and token count measure different properties; neither alone proves downstream quality.

Uzan et al. (2024) ; Meister (2026)

Structure, Integrity and Robustness

Metric	Definition	Evidence boundary
Command / number integrity	Fraction of reference lexemes with no internal token cut.	Does not require isolation as a standalone token.
Lexical boundary P/R/F1	Token endpoints against reference lexical endpoints.	A byte tokenizer can obtain high recall.
KaTeX crossing rate	Tokens partially overlapping a reference syntax span / all tokens.	A whole-formula token can obtain low crossing.
Parameter-boundary recall	pylatexenc group-content endpoints retained as token boundaries.	Independent implementation; syntax proxy, not semantic gold.
Robustness	Layout token variation; local symbol edit effects outside a 16-byte neighborhood.	The local edit is not mathematical equivalence.

Coverage is part of the result

Unicode integrity, parser coverage and probe counts accompany the scores. No formula disappears from the common test merely because a diagnostic parser cannot measure it.

Meister (2026) ; Jung et al. (2024)

Three New Review Rounds of the Retained Release

Round	Checks actually executed	Scope
1. Data / provenance	Reclassify every formula; verify file IDs, hash, license/origin rules and source-disjoint pools.	56,000 catalog records.
2. Public-encoder replay	Reload every current model; reconstruct original payload; replay length, reference crossings and parameter boundaries.	234,556 formula-result rows.
3. Independent aggregation	Recompute paper averages, integrity and fit-calibrated probability bits; check common dictionary and seven timing repeats.	19 configurations, 109 metric records.

Status

All three new reviews passed. The earlier v8 freeze/evaluation additionally passed eight protocol/result checks and 201 implementation tests.

Reorganization integrity

Selected checkpoints are byte-identical copies of the frozen v8 artifacts. Current results are the seven-method subset, with no new training or test-based parameter changes.

New-Paper Results: 1,024-ID Trained Vocabulary

Common test: 4,294 formulas from 27 new source papers.

System	T/B ↓	Cross% ↓	Scope% ↑	Cmd% ↑	Num% ↑	MiB/s ↑
Boundary AST (ours)	0.3900	18.43	64.36	99.76	97.41	29.68
Math-atom BPE (ours)	0.3943	19.26	62.83	99.76	97.41	10.20
Byte BPE	0.4121	25.51	67.62	62.28	97.43	8.59
SP Unigram	0.4912	22.34	68.64	2.49	98.08	23.42
Qwen2.5	0.5318	21.62	84.76	2.16	97.41	3.14
Llama 3	0.5279	21.74	84.76	2.16	99.28	8.94
o200k	0.5246	22.24	84.91	2.19	99.28	14.32

Reading the table

T/B is paper-macro tokens per UTF-8 byte. Crossing and scope recall use different external syntax references; improving one does not imply improving the other.

Comparison scope

The two proposed systems share a dictionary. Byte BPE and Unigram share the ID budget and allowed data. Qwen2.5, Llama 3 and o200k retain their full vocabularies.

New-Paper Results: 2,048-ID Trained Vocabulary

Common test: 4,294 formulas from 27 new source papers.

System	T/B ↓	Cross% ↓	Scope% ↑	Cmd% ↑	Num% ↑	MiB/s ↑
Boundary AST (ours)	0.3296	24.17	53.87	99.76	97.89	28.65
Math-atom BPE (ours)	0.3372	24.87	51.18	99.76	97.57	9.56
Byte BPE	0.3504	31.46	59.79	74.75	97.61	8.81
SP Unigram	0.4625	24.03	67.07	4.16	98.71	23.54
Qwen2.5	0.5318	21.62	84.76	2.16	97.41	3.14
Llama 3	0.5279	21.74	84.76	2.16	99.28	8.94
o200k	0.5246	22.24	84.91	2.19	99.28	14.32

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Strict Source-Domain Generalization

Common holdout: 1,428 formulas. Specialized training excludes target and adjacent source categories.

System	T/B: 1,024	T/B: 2,048	Scope%: 1,024	Scope%: 2,048
Boundary AST (ours)	0.4372	0.4010	61.27	53.43
Math-atom BPE (ours)	0.4470	0.4118	60.70	53.72
Byte BPE	0.4726	0.4188	66.29	61.13
SP Unigram	0.5325	0.5011	70.89	70.10
Qwen2.5	0.5415	0.5415	86.87	86.87
Llama 3	0.5353	0.5353	86.87	86.87
o200k	0.5335	0.5335	86.63	86.63

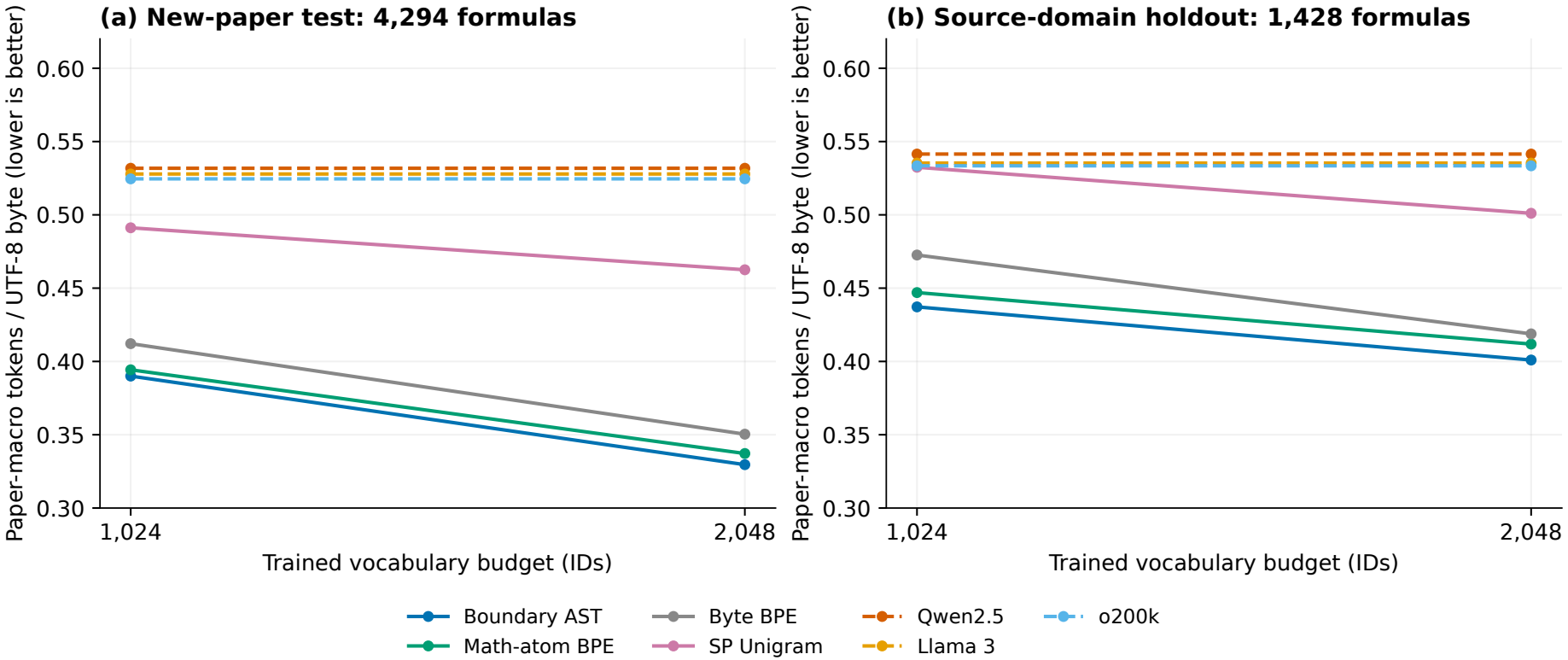
What generalizes

Boundary AST compresses more than the same-dictionary BPE in both budgets. Scope recall changes by +0.57 pp / -0.28 pp; not every structural diagnostic improves.

What remains uncertain

Source categories are disjoint by construction, but mathematical concepts can overlap. Few source papers and uneven domains produce wide uncertainty intervals.

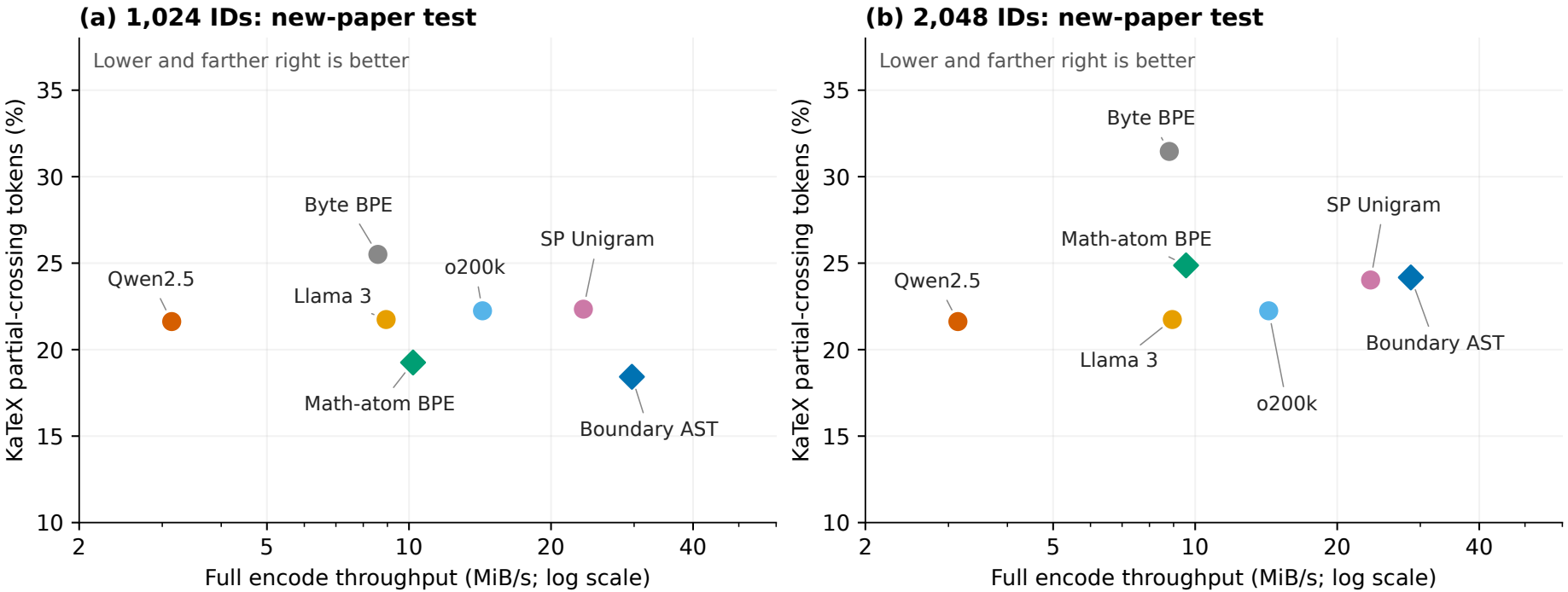
Compression Across Vocabulary Budgets



Interpretation

Solid lines: trained 1,024/2,048-ID systems. Dashed lines: fixed full-vocabulary general tokenizers. Lower is shorter; this is not a downstream-accuracy curve.

Runtime and Source-Tree Crossing



Interpretation

Boundary AST is about 2.65–3.00 times faster than math-atom BPE across the four principal conditions. All parser/codec work is included. The result is from one Mac and a single-input public interface.

Paired Paper-Cluster Statistical Evidence

Difference: Boundary AST minus math-atom BPE. Negative compression differences favor AST.

Condition	T/B difference	Family 95% CI	Scope difference CI (pp)
1,024 / broad	-0.00428	[-0.00601, -0.00270]	[-2.36, +6.06]
1,024 / domain	-0.00976	[-0.02916, -0.00197]	[-4.44, +9.88]
2,048 / broad	-0.00760	[-0.01106, -0.00535]	[+0.49, +6.59]
2,048 / domain	-0.01088	[-0.03025, -0.00344]	[-5.78, +12.17]

Design

200,000 paper-cluster resamples; Bonferroni intervals retain the original family of 144 comparisons. We did not shrink the family after choosing seven systems.

- All four adjusted compression intervals exclude zero in favor of Boundary AST.
- Scope and crossing intervals do not all exclude zero; point estimates are not universal structural gains.
- Intervals condition on the realized vocabulary and observed training seeds. They do not cover all possible retraining outcomes.

Probability Coding and Vocabulary Utilization

System	Fit BPB: 1,024	Fit BPB: 2,048	Used%: 1,024	Used%: 2,048
Boundary AST (ours)	3.131	3.008	67.3	72.0
Math-atom BPE (ours)	3.153	3.028	66.8	70.9
Byte BPE	3.413	3.209	76.3	76.3
SP Unigram	3.709	3.634	69.5	67.2
Qwen2.5	4.063	4.063	0.8	0.8
Llama 3	4.056	4.056	1.1	1.1
o200k	4.060	4.060	0.7	0.7

Interpretation

Fit BPB uses the same scenario fit records and actual vocabulary size. The much larger general vocabularies incur a different smoothing term and utilization denominator.

Why several measures are necessary

Low token count can coexist with sparse usage or unstable boundaries. These probability readouts are descriptive token statistics; no trained language-model perplexity is reported.

Uzan et al. (2024) ; Meister (2026)

Current Seven-System Composite Ranking

System	Rank: 1,024	Rank: 2,048	Score: 1,024	Score: 2,048
Boundary AST (ours)	1	1	0.042	0.100
Math-atom BPE (ours)	2	2	0.271	0.342
Byte BPE	5	5	0.610	0.608
SP Unigram	3	3	0.433	0.450
Qwen2.5	7	7	0.860	0.817
Llama 3	6	6	0.696	0.646
o200k	4	4	0.587	0.537

Frozen metric preferences

Compression 30%; encode speed 25%; KaTeX crossing 20%; command integrity 10%; number integrity 5%; fit BPB 10%. Broad and specialized scenarios each receive 50%.

How to use the ranking

The ranking favors our complete specialized systems under this preference. It is relative to the seven retained methods, omits scope recall from its weighted score and cannot establish downstream superiority.

Why RL Is Not a Final Production Route

Same-reward control	RL mean utility gain	Wins	95% interval crosses zero?
Fixed random search	0.116%	7 / 15	Yes
Cross-entropy search	0.199%	8 / 15	Yes

Controlled evidence

Three paper splits and five seeds per split; shared tasks, features, initialization and 34 actual terminal evaluations. The gate required at least 0.5% utility improvement, 12/15 wins and core noninferiority.

- The gate was not met; these results remain in history, outside the current seven-system registry.
- The non-RL Boundary AST replacement preserves core point-estimate tolerances relative to the old v5, with explicit structural trade-offs.
- This does not refute the ICML end-to-end RL paper, whose reward is language-model loss.

Historical v8 controlled-terminal decision; 15 pairs are conditional on three potentially overlapping source splits.

Dauncey and Wattenhofer (2026)

Four Priority Downstream Tasks

Task	Input and output	Primary evaluation
1. PDE solution / solver generation	Equation + initial/boundary conditions; generate an analytic candidate or executable numerical solver.	Residual, boundary error, nRMSE and execution success.
2. Governing-equation discovery	Field observations / derivatives; generate a symbolic PDE and fit coefficients.	Structure recovery, coefficient error, held-out prediction and noise robustness.
3. Symbolic manipulation	Integration, differentiation, simplification and ODE solving.	Executable/symbolic equivalence, validity and unseen operator combinations.
4. Mathematical reasoning	Problem text + formulas; generate derivation and final answer.	Verified final-answer accuracy and length/generalization strata.

Data and task separation

PDEBench supplies numerical PDE tasks; EqGPT motivates discovery; SymbolicMathematics and MATH motivate symbolic/reasoning tasks. Their test cases must not enter tokenizer or model adaptation training.

Takamoto et al. (2022) ; Xu et al. (2025) ; Lample and Charton (2020) ; Hendrycks et al. (2021)

Tokenizer Replacement Does Not Require Full Retraining

Option	Procedure	Experimental use
Small models from scratch	Same backbone, raw data and initialization seeds.	Clean comparison; start around 100–300M parameters.
Adapt an open backbone	Replace vocabulary; initialize embeddings; continue pretraining.	Retain a no-replacement adaptation control.
Align / transplant embeddings	Use old-token decompositions or learned alignment.	Equal adaptation cost against simple initialization.

What must change

New IDs cannot reuse old embedding rows by ID. Replace input/output vocabulary matrices; other Transformer weights can be retained.

Core difficulty

Formula-only data are insufficient for reasoning and solver generation. Additional context/task data and mixed-text coverage are needed.

Li, Zhang and Zong (2025) ; Limisiewicz et al. (2026)

Compute, Vocabulary Cost and Fair Training Budgets

Vocabulary-dependent parameters

Tied embeddings: $P_{\text{vocab}} = |V|d$. At $d = 2{,}048$:
$|V| = 2{,}048 \Rightarrow 4.2$M; $|V| \approx 200{,}000 \Rightarrow 409.6$M parameters. Untied input/output matrices roughly double this component.

Budget issue	Protocol requirement
Training amount	Use the same raw examples/bytes; also report tokens and measured FLOPs.
Context length	Report raw-byte coverage and truncation, not only a common token window.
Model size	Report non-embedding and total parameters separately.
Compute scale	Heuristic $C \approx 6ND$: a 1B-parameter, 10B-token run is about 6×10^{19} FLOPs before additional overhead.
Replication	Seven systems times three seeds is 21 runs per task configuration.

Staged resource plan

Start with equal-capacity small-model tests. Scale after a reproducible task gain; cost full-vocabulary deployment references separately.

Limisiewicz et al. (2026)

Implementation Sequence and Decision Criteria

Phase	Work	Advance only if
A. Freeze task protocol	Choose equation families, source splits, validation rules and resource accounting.	Data leakage and task evaluator checks pass.
B. Small-model pilot	Two routes + five baselines; matched raw training data, same task code, three seeds.	Gain is stable under task accuracy and compute accounting.
C. Open-model adaptation	Token/embedding initialization ablations with equal adaptation budgets.	Benefits exceed the no-replacement control.
D. Generalization / scale	New PDE families, long formulas, noise and mixed text/formula prompts.	Gain persists across distributions without unacceptable regressions.

Failure conditions to investigate

No task gain despite better compression; invalid expression generation; failure on unseen domains; adaptation cost larger than inference savings; arithmetic or scope errors hidden by string similarity.

Stage 2: Autonomous and Signal-Guided Tokenization

Research objective

Learn boundaries with little manual intervention and excellent quality, generalization and full-interface speed.

Direction	Learning signal and mechanism	Question to resolve
Fast signal guidance	Predictive uncertainty or learned structure; distill a fast encoder.	Which signal improves quality after runtime cost?
Autonomous boundaries	Causal prediction feedback; RL versus equal-budget search.	Do gains survive deployment with fewer rules?
Mathematical transitions	Predict verified operations; learn units without boundary labels.	Do units transfer to unseen operations and compositions?

Current decision

Stage 2 is paused as future research. No system has passed both quality and speed requirements; mathematical understanding remains unproven.

Dauncey and Wattenhofer (2026) ; Hwang, Wang and Gu (2025) ; Lample and Charton (2020)

Research Priorities and Acceptance Criteria

Priority	Planned experiment	Required evidence
Now: Stage 1 tasks	Two frozen routes and five baselines; common task data and budgets.	Verified symbolic/PDE results; fair compute accounting.
Later: learned signals	Signal ablations, fast distillation, zero/shuffled controls.	Signal-specific quality gains and full encode speed.
Later: autonomous learning	Longer causal context, joint training, consistent deployment.	Stable gains over simple controls; fewer manual rules.
Later: mathematical laws	Verified operations; source and composition holdouts.	Correctness beyond the training environment.

Two conditions must hold together

Substantially reduced human intervention and excellent multi-metric performance. Prediction proxies alone are insufficient; disclose remaining priors and regressions.

Research handoff

Preserve Stage 2 experiments for later work. Proceed with the two Stage 1 routes and five baseline tokenizers.

What We Can Claim, and What We Must Test Next

Supported by current evidence

A classified, source-traceable formula catalog and reproducible frozen experiments are available. Boundary AST improves compression over a same-dictionary BPE control while offering a useful speed/structure trade-off.

Supported statement	Boundary
Two routes are deployable	Standalone tokenizers only; no downstream model training has been completed.
Three current audits passed	Automated data, implementation and experiment checks; not external certification.
Some core metrics improve	Not all structure metrics or adjusted intervals favor one method.
Specialization is promising	Sparse PDE coverage, uneven domains and mixed-text transfer remain open.

Next scientific question

Do the observed intrinsic improvements survive embedding adaptation and produce better symbolic/PDE task performance at a defensible total training cost?

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